

ASSIP 2026 — Weekly Assignment

Mentor: Prof. Lei Gao (George Mason University) · Week of July 10, 2026

Two parts this week. **Part 1** is a written research brief. **Part 2** prepares you to discuss a real finance paper with me. Read both fully before you start.

Part 1 — The Other Side of the “AI Takes Your Job” Story

Due: Monday, July 13.

The headlines say “*AI is coming for your job.*” That is only half the story. A large and growing body of serious research finds that AI often **complements** human work — raising productivity, and sometimes helping the people who need it most. Your task is to investigate that other half, with special attention to **workers with disabilities**, for whom AI-powered tools (speech-to-text, screen readers, live captioning, AI job-matching, remote-work software) can open doors that were previously closed.

What to produce — a 2–3 page research brief with four parts

1. **Annotated bibliography** of **5–7 credible sources** (at least **2** specifically on disability & employment). For each source, write 2–3 sentences: what did it find, and why does it matter?
2. **Two datasets** you could actually use. For each, state **one specific research question** you could answer with it.
3. **Your own hypothesis** — one sentence — plus a short paragraph sketching how you would test it (what data, what comparison).
4. **One honest caveat.** The evidence is genuinely mixed. Find and describe at least one way AI can *harm* disabled workers (e.g., biased hiring tools). **Good research acknowledges the other side.**

Start here — then go beyond these

Positive effects of AI on work (augmentation, productivity): - Brynjolfsson, Li & Raymond (2023), *Generative AI at Work*, NBER 31161 — an AI assistant raised customer-support productivity 14% on average, and **34% for the least-experienced workers.** : <https://www.nber.org/papers/w31161> - Noy & Zhang (2023), *Science* — in an experiment, Chat-GPT cut writing-task time ~40%, raised quality, and **narrowed the gap between weaker and stronger writers.** : <https://pubmed.ncbi.nlm.nih.gov/37440646/> (open preprint available on SSRN) - Autor (2024), *Applying AI to Rebuild Middle Class Jobs*, NBER 32140 — argues AI can extend human expertise to a broader set of workers. : <https://www.nber.org/papers/w32140>

AI, assistive technology & disability employment: - Kruse, Schur et al. (2024), *Assistive Technology’s Potential to Improve Employment of People with Disabilities*, J. Occupational Rehabilitation — U.S. Census data links greater assistive-technology availability to **higher disability employment.** : <https://pmc.ncbi.nlm.nih.gov/articles/PMC11219186/> - Schur, Ameri & Kruse (2020), *Telework after COVID: a “silver lining” for workers with disabilities?* — remote work, enabled by digital/assistive tools, removes commuting and physical-workplace barriers. : <https://smlr.rutgers.edu/faculty-research-engagement/program-disability-research/telework> - OECD (2023), *Using AI to Support People with Disability in the Labour*

Market — four categories of AI that help disabled workers, from 70+ stakeholder interviews.
: https://www.oecd.org/en/publications/using-ai-to-support-people-with-disability-in-the-labour-market_008b32b7-en.html

The honest nuance (AI can also harm — you must engage with this): - ILO (2025), *AI's double-edged sword* — the *same* AI can discriminate: resume screeners that downrank disability-related signals; video-interview tools that misread autism or facial differences. : <https://www.ilo.org/resource/article/ais-double-edged-sword-new-frontier-employment-people-disabilities>

Datasets to explore

- **IPUMS CPS** — U.S. monthly labor-force data with a disability question (DIFFANY, since 2008). Free with registration; build custom extracts online. : <https://cps.ipums.org/cps/>
- **DisabilityStatistics.org** (Cornell University) — **no coding required:** pre-made tables of disability employment rates by year and state. Great first stop. : <https://www.disabilitystatistics.org/>
- **AI Occupational Exposure (AIOE)** — which occupations are most “exposed” to AI. Free on GitHub. : <https://github.com/AIOE-Data/AIOE>
- ****O*NET**** — detailed attributes of ~1,000 occupations (skills, tasks, work context). Free. : <https://www.onetcenter.org/database.html>
- *Also useful:* **nTIDE** monthly disability-employment report (<https://researchondisability.org/nTIDE>); **IPUMS ACS** for occupation × disability (<https://usa.ipums.org/usa/>).

Ground rules (important)

- **Every citation must be real and clickable — verify each one yourself by opening the link.** Do **not** trust any AI-generated citation without checking it. Fabricated references are the fastest way to lose credibility in research.
- Prefer the **primary source** (the actual paper) over a blog summary.
- Keep a running list of the search terms and databases you used — you’ll include it at the end.

Guiding questions (pick a few to anchor your brief)

- Does AI help *lower-skilled* workers more than higher-skilled ones? Why might that be?
- Does remote work (enabled by AI/assistive tech) expand employment for disabled workers?
- Which occupations are most “exposed” to AI — and is *exposure* the same thing as *at risk*?

Part 2 — Read One of My Papers

Discussion: Tuesday, July 14 (morning) — come ready to talk for 5 minutes.

Pick **one** paper below, read it carefully, and prepare a **5-minute talk**. This is how you learn to read finance research. You do **not** need to understand every equation — focus on the story.

In your 5 minutes, cover these five things (this is also how every finance paper is built)

1. **The question** — what are they asking, and why does it matter?
2. **The data** — what data did they use, and where did it come from?
3. **The method** — how do they answer it? (What is the key comparison or “experiment”?)
4. **The main finding** — the single most important result, *with the number*.
5. **Your take** — one thing you found clever, and one thing you would question or want to know more.

Choose one (all are mine; roughly easiest : most technical)

1. **Lending Practices to Same-Sex Borrowers** (Sun & Gao, *PNAS* 2019) — *Do same-sex couples face mortgage discrimination?* Uses real federal mortgage records. Intuitive and socially important (ranked in the *Financial Times* Top-100 research with social impact). **Best first read.**
2. **The Rainbow of Municipal Credit: Evidence from Same-Sex Marriage Bans** (Gao, Liu & Wang, working paper) — extends the discrimination question to the *municipal bond* market: do same-sex-marriage bans raise a city’s borrowing costs? Relatable policy question.
3. **Does Local Religiosity Affect Organizational Risk-Taking? Evidence from the Hedge Fund Industry** (Gao, Wang & Zhao, *J. Corporate Finance* 2017) — *Do hedge funds in more religious areas take less risk?* Textbook-clear question : data : method : finding.
4. **Local Religiosity and Financial Advisor Misconduct** (Gao, Cowan, Han & Pan, *J. Corporate Finance* 2024) — the same “does local culture shape behavior?” idea applied to individual financial advisers and their misconduct records. Pairs nicely with paper #3.
5. **Corporate Patents, R&D Success, and Tax Avoidance** (Gao, Yang & Zhang, *Rev. Quant. Finance & Accounting* 2016) — *Do successful innovators also avoid more tax?* (Nice fit if you’re on one of the patents projects.)
6. **Standing Out from the Crowd via CSR Engagement** (Gao, He & Wu, *J. Financial & Quantitative Analysis* 2023) — *Why do firms “do good”?* A competition-based answer. (ESG.)
7. **Individualistic CEO and Corporate Innovation** (Gao, Han, Pan & Zhang, *Research Policy* 2023) — *Does a CEO’s cultural background (U.S. frontier heritage) shape how much the firm innovates?*
8. **Seeing is Believing: Annual Report “Graphicity” and Stock Return Predictability** (Gao, Deng, Hu & Zhou, working paper) — *Do the pictures and charts in a company’s annual report carry information the market misses?* A modern, novel angle (text- and image-as-data).
9. **Market Intraday Momentum** (Gao, Han, Li & Zhou, *J. Financial Economics* 2018) — *Does the market’s morning return predict its afternoon return?* A simple, tradable idea. **Most technical / for the quant-minded.**

How to access

- My **Google Scholar** page has a free PDF or link for every paper:
<https://scholar.google.com/citations?user=zK0dvhsAAAAJ>
- Or search “**Lei Gao George Mason**” on SSRN (<https://www.ssrn.com>).

How to read your first finance paper (tips)

- Read in this order: **abstract : introduction : conclusion : then the ONE main table.** Skip the heavy math on the first pass.
- The **introduction previews every finding** — read it twice.
- Write down the **one sentence** that captures the whole paper. If you can say it out loud, you understand it.

Questions during the week? Email me. I'd rather you ask early than stay stuck.